

화학수학

여덟 번째 수업

함수급수

Sequence and Series

- **Sequence** (수열): $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_n \rightarrow \cdots$
 - Rule for generating one member from previous one
- **Series** (급수)
 - Finite series (유한급수): $s = a_0 + a_1 + \cdots + a_n$
 - Infinite series (무한급수): $s = a_0 + a_1 + \cdots + a_n + \cdots$

$$s = a_0 + a_1 + \cdots + a_n + \cdots = \sum_{k=0}^{\infty} a_k$$

Geometric Series (기하급수)

$$s = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = a \sum_{n=0}^{\infty} r^n$$

- Convergence condition: $|r| < 1$
- Value: since $s = a + r(a + ar + ar^2 + \cdots) = a + rs$,

$$s = \frac{a}{1 - r}$$

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

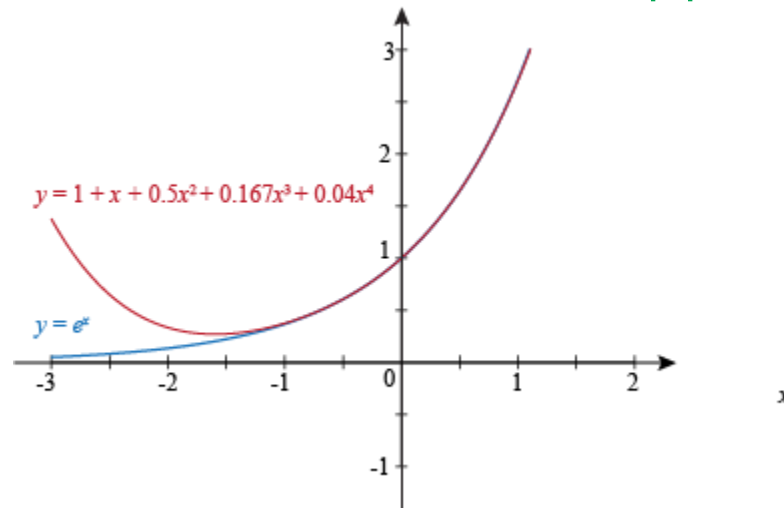
- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Approximation to a Function

- Functional series can be used to represent a function

$$s(x) = a_0g_0(x) + a_1g_1(x) + a_2g_2(x) + \cdots \rightarrow f(x)$$

- Its partial sum can be used as an approximation



(Image: <https://www.shmoop.com/series/taylor-maclaurin-series-examples.html>)

Maclaurin Series to Represent $f(x)$

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

At $x = 0$, $f(0) = a_0$ and

$$\left(\frac{d^n f}{dx^n} \right)_{x=0} = n! a_n$$

Hence,

$$a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

Taylor Series to Represent $f(x)$

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

where $h \neq 0$.

$$a_0 = f(h), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=h}$$

- Often called **Taylor expansion about $x = h$**
- A function that is not differentiable at $x = 0$ can be represented by a Taylor series

Change of Variable

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

can be written as

$$f(x + h) = a_0 + a_1x + a_2x^2 + \dots$$

$$a_0 = f(x = 0), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

e.g. Taylor expansion of $\ln(x + 1)$:

$$\ln(x + 1) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

Mathematical Operations on Series

Differentiation/integration of a series?

For a (uniformly convergent) functional series

$$f(x) = \sum_{n=0}^{\infty} a_n g_n(x),$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] = \sum_{n=0}^{\infty} a_n \frac{dg_n(x)}{dx}$$

$$\int_b^c f(x) dx = \int_b^c \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] dx = \sum_{n=0}^{\infty} a_n \int_b^c g_n(x) dx$$

Today's Class

- Fourier series
 - Sinusoidal basis functions
 - Complex exponential basis functions
- General functional series
 - Hilbert space
 - Scalar product, orthogonality, and normalization
 - Determination of coefficients

11. Functional Series and Integral Transforms

(11장 함수급수와 적분변환)

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

- **Periodicity:** $f(x) = f(x + 2L) = f(x + 4L) = \dots$
- **Completeness:** the series can exactly represent any function

Orthogonality

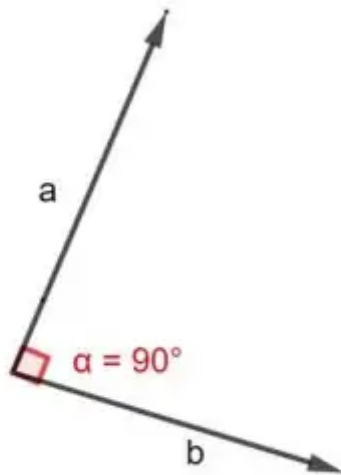
The basis functions are orthogonal to each other:

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 2L, & \text{if } m = n = 0 \\ L, & \text{if } m = n \neq 0 \\ 0, & \text{if } m \neq n \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m = n = 0 \\ L, & \text{if } m = n \neq 0 \\ 0, & \text{if } m \neq n \end{cases}$$

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

Vector Orthogonality



$$\vec{a} \perp \vec{b}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \alpha = 0$$

(Image: <https://www.storyofmathematics.com/orthogona-vector>)

Kronecker Delta

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

We can rewrite the orthogonality conditions:

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 2L, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

Multiplying both sides by $\cos(m\pi x/L)$ and integrating,

$$\begin{aligned} & \int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx \\ &= \sum_{n=0}^{\infty} a_n \int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\ &+ \sum_{n=0}^{\infty} b_n \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \end{aligned}$$

0

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = \sum_{n=0}^{\infty} a_n \int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx$$

Because of orthogonality, only the terms with $n = m$ survive:

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = a_m \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx$$

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = a_m \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx$$

For $m = 0$,

$$\int_{-L}^L f(x) dx = a_0 \times (2L)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = a_m \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx$$

For $m \neq 0$,

$$\int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx = a_m \times L$$

$$a_{m \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx$$

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

Multiplying both sides by $\sin(m\pi x/L)$ and integrating,

$$\begin{aligned} & \int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx \\ &= \sum_{n=0}^{\infty} a_n \int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \\ &+ \sum_{n=0}^{\infty} b_n \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \end{aligned}$$

0

Finding the Coefficients

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx = \sum_{n=0}^{\infty} b_n \int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx$$

Similarly, we obtain

$$b_m = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

$$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx$$

Odd and Even Functions (홀함수와 짝함수)

- Odd function:

$$f(-x) = -f(x)$$

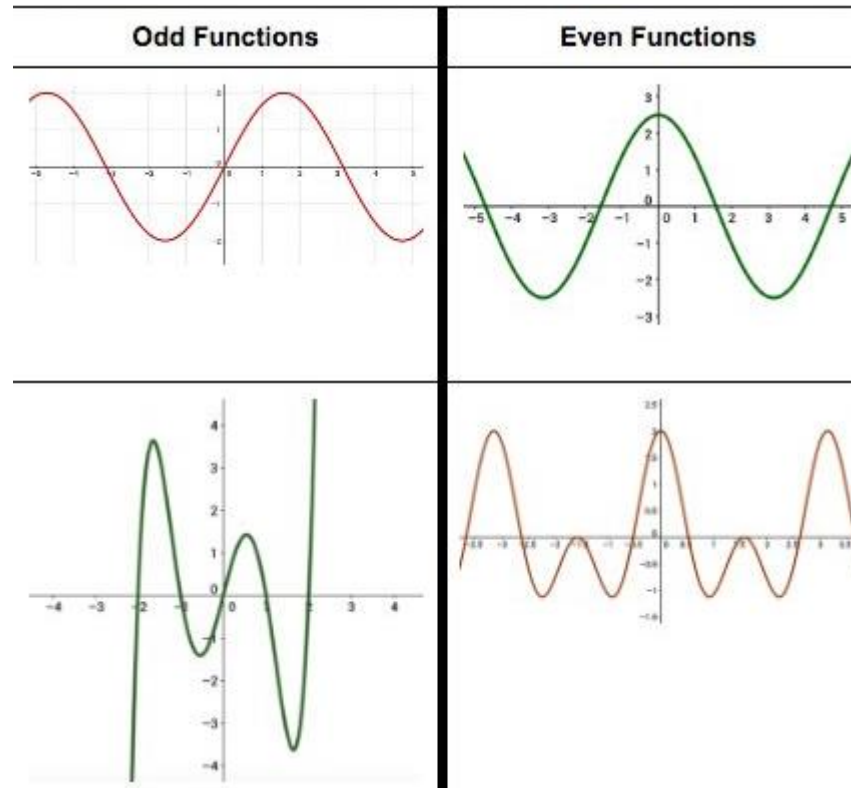
e.g. $\sin x$

- Even function:

$$f(-x) = f(x)$$

e.g. $\cos x$

Odd and Even Functions



Odd and Even Functions

$$(\text{Odd}) \times (\text{Odd}) = (\text{Even})$$

$$(\text{Odd}) \times (\text{Even}) = (\text{Odd})$$

$$(\text{Even}) \times (\text{Even}) = (\text{Even})$$

For an odd function $f(x)$,

$$\int_{-c}^c f(x) \, dx = 0$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

	Odd $f(x)$	Even $f(x)$
$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$	0	
$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$	0	
$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$		0

sine series cosine series

Example 11.1

Find the Fourier series to represent the function $f(x) = x$ for the interval $-L < x < L$.

The function is odd, and the series will be a sine series:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{1}{L} \int_{-L}^L x \sin\left(\frac{n\pi x}{L}\right) dx$$

Example 11.1

Find the Fourier series to represent the function $f(x) = x$ for the interval $-L < x < L$.

$$b_n = \frac{1}{L} \int_{-L}^L x \sin\left(\frac{n\pi x}{L}\right) dx$$

By defining $y = n\pi x/L$,

$$dy = (n\pi/L)dx$$

$$x = -L \rightarrow y = -n\pi$$

$$x = L \rightarrow y = n\pi$$

$$b_n = \frac{1}{L} \left(\frac{L}{n\pi}\right)^2 \int_{-n\pi}^{n\pi} y \sin y dy$$

Example 11.1

Find the Fourier series to represent the function $f(x) = x$ for the interval $-L < x < L$.

$$b_n = \frac{1}{L} \left(\frac{L}{n\pi} \right)^2 \int_{-n\pi}^{n\pi} y \sin y \, dy$$

Setting $u(y) = y$ and $dv/dy = \sin y$ (or $v = -\cos y$),

$$\begin{aligned} b_n &= \frac{L}{(n\pi)^2} \int_{-n\pi}^{n\pi} u \frac{dv}{dy} dy = \frac{L}{(n\pi)^2} \left[uv \Big|_{-n\pi}^{n\pi} - \int_{-n\pi}^{n\pi} \frac{du}{dy} v dy \right] \\ &= \frac{L}{(n\pi)^2} \left[-y \cos y \Big|_{-n\pi}^{n\pi} + \int_{-n\pi}^{n\pi} \cos y \, dy \right] = \frac{L}{(n\pi)^2} \left[-y \cos y \Big|_{-n\pi}^{n\pi} + \sin y \Big|_{-n\pi}^{n\pi} \right] \\ &= \frac{L}{(n\pi)^2} [-(n\pi) \cos(n\pi) + (-n\pi) \cos(-n\pi)] = \frac{2L}{n\pi} (-1)^{n-1} \end{aligned}$$

Example 11.1

Find the Fourier series to represent the function $f(x) = x$ for the interval $-L < x < L$.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$
$$b_n = \frac{2L}{n\pi} (-1)^{n-1}$$

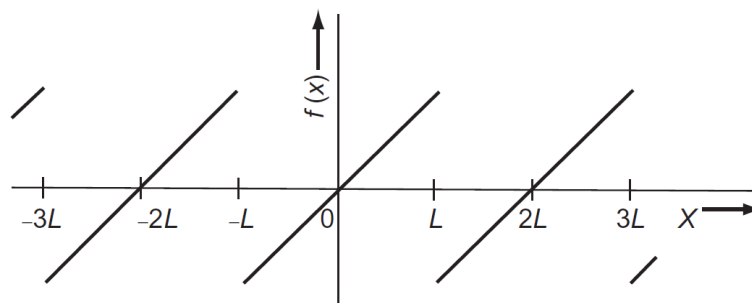
Therefore, the series is

$$f(x) = \sum_{n=1}^{\infty} \frac{2L}{n\pi} (-1)^{n-1} \sin\left(\frac{n\pi x}{L}\right)$$

Example 11.1

Find the Fourier series to represent the function $f(x) = x$ for the interval $-L < x < L$.

$$f(x) = \sum_{n=1}^{\infty} \frac{2L}{n\pi} (-1)^{n-1} \sin\left(\frac{n\pi x}{L}\right)$$

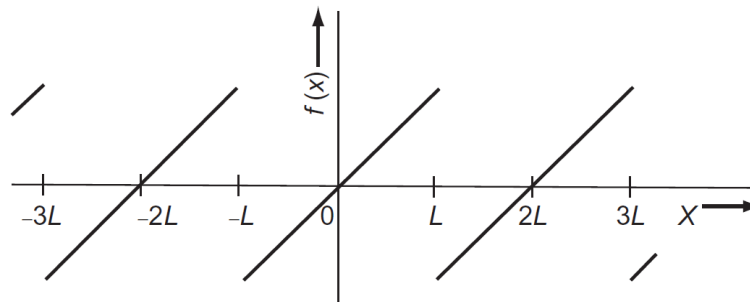


(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 11.1)

Example 11.2

Show that the series of the previous example equals zero at $x = -L$, $x = L$, and $x = 3L$, the value halfway between the two values of the function at the step discontinuity.

$$f(x) = \sum_{n=1}^{\infty} \frac{2L}{n\pi} (-1)^{n-1} \sin\left(\frac{n\pi x}{L}\right)$$



Example 11.2

Show that the series of the previous example equals zero at $x = -L$, $x = L$, and $x = 3L$, the value halfway between the two values of the function at the step discontinuity.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

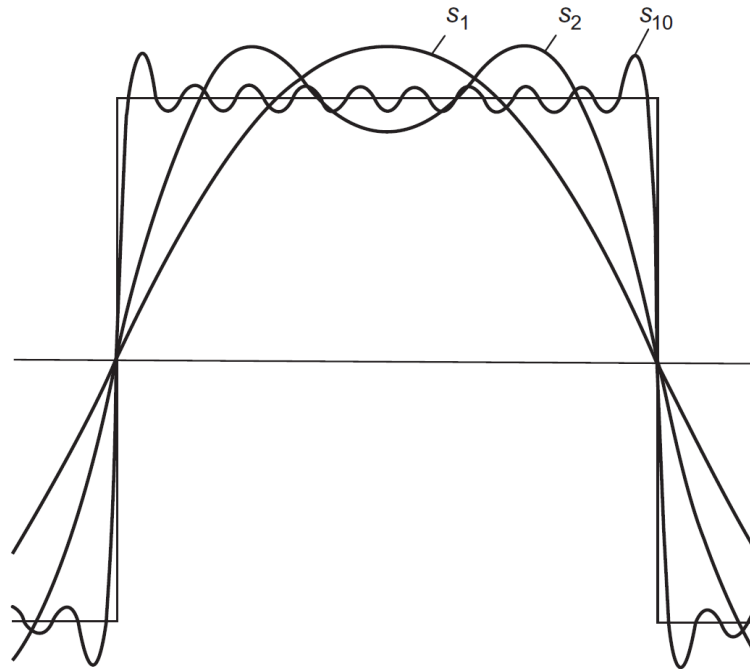
$$f(-L) = \sum_{n=1}^{\infty} b_n \sin(-n\pi) = 0$$

$$f(L) = \sum_{n=1}^{\infty} b_n \sin(n\pi) = 0$$

$$f(3L) = \sum_{n=1}^{\infty} b_n \sin(3n\pi) = 0$$

Approximation by Partial Sum

$$f(x) = \begin{cases} +1 & \text{for } 0 < x < L \\ -1 & \text{for } -L < x < 0 \end{cases}$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 11.2)

Complex Exponential Basis Functions

$$\cos \alpha = \frac{e^{i\alpha} + e^{-i\alpha}}{2}, \quad \sin \alpha = \frac{e^{i\alpha} - e^{-i\alpha}}{2i}$$

Thus,

$$\begin{aligned} & a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \\ &= \frac{a_n}{2} (e^{in\pi x/L} + e^{-in\pi x/L}) + \frac{b_n}{2i} (e^{in\pi x/L} - e^{-in\pi x/L}) \\ &= \frac{a_n - ib_n}{2} e^{in\pi x/L} + \frac{a_n + ib_n}{2} e^{-in\pi x/L} \end{aligned}$$

Complex Exponential Basis Functions

$$a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{a_n - ib_n}{2} e^{in\pi x/L} + \frac{a_n + ib_n}{2} e^{-in\pi x/L}$$

The following two formulas are equivalent:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

Orthogonality Relation

$$\int_{-L}^L \left(e^{im\pi x/L}\right)^* e^{in\pi x/L} dx = 2L\delta_{mn}$$

Finding the Coefficients

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

Multiplying both sides by $e^{-im\pi x/L}$ and integrating,

$$\int_{-L}^L f(x) e^{-im\pi x/L} dx = \sum_{n=-\infty}^{\infty} c_n \int_{-L}^L e^{-im\pi x/L} e^{in\pi x/L} dx$$

From the orthogonality relation,

$$\int_{-L}^L f(x) e^{-im\pi x/L} dx = \sum_{n=-\infty}^{\infty} c_n (2L\delta_{mn}) = 2Lc_m$$

Finding the Coefficients

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

$$\int_{-L}^L f(x) e^{-im\pi x/L} dx = \sum_{n=-\infty}^{\infty} c_n (2L\delta_{mn}) = 2Lc_m$$

Therefore,

$$c_m = \frac{1}{2L} \int_{-L}^L f(x) e^{-im\pi x/L} dx$$

General Functional Series

- Complete set of orthogonal basis functions:

$$\psi_1, \psi_2, \dots, \psi_n, \dots$$

- Expansion of an arbitrary function f :

$$f = \sum_n c_n \psi_n$$

- Hilbert space: $\psi_1, \psi_2, \dots, \psi_n, \dots$ span the Hilbert space

Scalar Products in the Hilbert Space

- The scalar product of $f(x)$ and $g(x)$ is

$$\int f^*(x)g(x) dx$$

- The scalar product of $f(x, y, z)$ and $g(x, y, z)$ is

$$\iiint f^*(x, y, z)g(x, y, z) dx dy dz$$

Orthogonality and Normalization

- The basis functions are **orthogonal**:

$$\int \psi_n^* \psi_m \, dx = c \delta_{nm}$$

- They can be **normalized**:

$$\int \psi_n^* \psi_m \, dx = \delta_{nm}$$

- We assume the normalized basis functions

Finding the Coefficients

$$f(x) = \sum_n c_n \psi_n$$

Multiplying both sides by ψ_m^* and integrating,

$$\int \psi_m^* f dx = \sum_n c_n \int \psi_m^* \psi_n dx$$

$$\int \psi_m^* f dx = \sum_n c_n \delta_{mn} = c_m$$

Example 11.3

The normalized quantum-mechanical wave functions for the particle in a one-dimensional box of length a are

$$\psi_n = \left(\frac{2}{a}\right)^{1/2} \sin\left(\frac{n\pi x}{a}\right)$$

These functions are a complete set for expansion of functions that are defined in the region $0 < x < a$ and vanish at $x = 0$ and at $x = a$. Find the coefficient c_1 if $f = x^2 - ax$.

We utilize the following formula:

$$c_n = \int \psi_n^* f \, dx$$

Example 11.3

$$\psi_n = \left(\frac{2}{a}\right)^{1/2} \sin\left(\frac{n\pi x}{a}\right), \quad f = x^2 - ax$$

$$c_1 = \int \psi_1^* f \, dx = \int_0^a \left(\frac{2}{a}\right)^{1/2} \sin\left(\frac{n\pi x}{a}\right) \times (x^2 - ax) \, dx$$

$$c_1 = \left(\frac{2}{a}\right)^{1/2} \int_0^a x^2 \sin\left(\frac{n\pi x}{a}\right) \, dx - (2a)^{1/2} \int_0^a x \sin\left(\frac{n\pi x}{a}\right) \, dx$$

Using the table of integrals,

$$c_1 = \left(\frac{2}{a}\right)^{1/2} \left(\frac{a}{\pi}\right)^3 (\pi^2 - 4) - (2a)^{1/2} \left(\frac{a}{\pi}\right)^2 \pi = -\frac{4\sqrt{2}a^{5/3}}{\pi^3}$$